

THE TRANSPORT CONJECTURE BETWEEN LATTICE $SU(N)$ GLUEBALLS AND BIANCHI ORBIFOLD SPECTRAL OBSERVABLES: FORMAL STATEMENT, SHARPENED FALSIFIER, TWO-MILLENNIUM-STACK SCOPING

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ABSTRACT. We formulate (V2) an open conjectural identification, the *Transport Conjecture*, between (i) the lattice $SU(N)$ scalar glueball mass $m_{0++}(SU(N))$, and (ii) a spectral observable $m_{\text{spectral}}^{(\chi_*)}$ on a Bianchi 3-orbifold Y_K with $K = K_{\text{ASP}}(N)$ a privileged imaginary quadratic anchor specified by the Center–Rank theorem of [R26a]. Relative to V1 [R26e], we (a) make the $K_{\text{ASP}}(N)$ map explicit per the Center–Rank canonical table (Table 1), (b) replace the primary falsifier by the cleaner Boolean reformulation $\lambda_1 = 1$ (no exceptional discrete eigenvalue below the continuous spectrum), (c) state explicitly that the Transport Conjecture is a *2-Millennium-level stack* (4D YM constructive existence + spectral identification Φ), and (d) refine the tier classification: the conjecture remains TIER 3 OPEN at uniform N , but for $N = 3$ the falsifier is operationally TIER 2 NUM (compute λ_1 of $\text{PSL}_2(\mathcal{O}_{\mathbb{Q}(\sqrt{-67})})$ via PARI/GP Selberg trace, budget $\leq \$50$ KVM compute, ≤ 3 months). Five falsifiable predictions covering $N \in \{2, 3, 4, 5, 6\}$ are stated explicitly. We add no new arXiv references over V1; all citations are verified against the published literature.

1. THE TRANSPORT CONJECTURE (V2 STATEMENT)

1.1. Setup. Let $N \geq 2$ be an integer and let $\mathbb{T}^4 = (\mathbb{R}/L\mathbb{Z})^4$ be the 4-torus of edge L . Let $\mathcal{A}_{SU(N)}(\mathbb{T}^4)$ denote the lattice $SU(N)$ Yang–Mills theory on $\mathbb{T}^4$ with Wilson action (or any of its standard improvements), and let $\mathcal{O}^{(0++)} : \mathcal{A}_{SU(N)}(\mathbb{T}^4) \rightarrow \mathbb{R}$ denote the lowest scalar glueball observable. In the continuum limit (lattice spacing $a \rightarrow 0$ at fixed physical volume), one defines

$$m_{0++}^{\text{lat}}(SU(N)) := \lim_{a \rightarrow 0} \sqrt{a} \cdot \langle \mathcal{O}^{(0++)} \hat{\mathcal{O}}^{(0++)} \rangle_a^{1/2}, \quad (1)$$

where the $\hat{\cdot}$ denotes a temporal-correlator extraction and $\langle \cdot \rangle_a$ the lattice ensemble average at spacing a .

On the arithmetic side, let $K = \mathbb{Q}(\sqrt{D})$ be imaginary quadratic with fundamental discriminant $D < 0$, $\mathcal{O}_K$ its ring of integers, $Y_K := \text{PSL}_2(\mathcal{O}_K) \backslash \mathbb{H}^3$ the associated Bianchi 3-orbifold [EGM98, §1.5], and Δ the positive hyperbolic Laplacian on Y_K . By the per-character spectral decomposition of the companion paper [R26b],

$$L^2(Y_K) = \bigoplus_{\chi \in \hat{g}(K)} L^2(Y_K)_\chi,$$

and $\Delta_\chi := \Delta|_{L^2(Y_K)_\chi}$ has its own discrete cuspidal + continuous Eisenstein spectrum (continuous spectrum supported on $[1, \infty)$ in our normalisation [EGM98, §§6.1, 7.4]). The Center–Rank

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ORCID 0009-0008-2443-7166. This work is part of the *crossed-cosmos* project (concept DOI 10.5281/zenodo.19686398). **V2 changes vs V1:** (i) explicit $K_{\text{ASP}}(N)$ table per Center–Rank canonical Cl-anchor map, (ii) primary falsifier reframed as the Boolean “no exceptional discrete eigenvalue” $\lambda_1 = 1$, (iii) explicit 2-Millennium-stack obstruction structure in §3, (iv) honest tier classification: TIER 2 NUM falsifier at $N = 3$, TIER 3 OPEN at uniform N .

theorem CR of [R26a] selects a privileged genus character χ_* for the anchor $K_{\text{ASP}}(N)$. Define

$$m_{\text{spectral}}^{(\chi_*)}(K, N) := \sqrt{\lambda_1(\Delta_{\chi_*})}, \quad (2)$$

the square root of the lowest cuspidal eigenvalue of Δ_{χ_*} on Y_K .

1.2. The $K_{\text{ASP}}(N)$ canonical anchor table (V2 explicit, D1 resolution). The $K_{\text{ASP}}(N)$ map is specified by the Center–Rank theorem CR of [R26a] as the unique-up-to-conjugation field with minimal $|D|$ satisfying $\text{rk}_2(\text{Cl}(K)) \geq v_2(N)$, with priority to Heegner ($h_K = 1$) when $v_2(N) = 0$:

N	$v_2(N)$	$K_{\text{ASP}}(N) = \mathbb{Q}(\sqrt{D})$	$\text{rk}_2(\text{Cl})$	Notes
2	1	$D = -15$	1	$h_K = 2$, smallest non-Heegner $\text{rk}_2 \geq 1$
3	0	$D = -67$	0	Heegner, smallest among class-number-1 anchors
4	2	$D = -84$	2	Klein V_4 , demotes V15 $D = -163$
5	0	$D = -19$	0	Heegner, demotes V16 $D = -15$
6	1	$D = -87$	1	consistent V15
7	0	$D = -43$	0	Heegner, demotes V16 $D = -420$
8	3	$D = -420$	3	consistent V15
9	0	$D = -163$	0	Heegner
≥ 10	various	various	various	per CR S1–S3 (Center–Rank)

TABLE 1. $K_{\text{ASP}}(N)$ canonical table per the Center–Rank theorem CR. The map is determined by the requirement $\text{rk}_2(\text{Cl}(K_{\text{ASP}}(N))) \geq v_2(N)$ and minimal $|D|$ (with Heegner priority at $v_2(N) = 0$). *V2 resolves the D1 divergence of the wave: workers occasionally used $D = -15$ for $N = 3$ (the smallest non-trivial $\text{rk}_2 \geq 1$ field), but the canonical $K_{\text{ASP}}(3) = \mathbb{Q}(\sqrt{-67})$ per CR.*

1.3. Transport Conjecture (V2 formal statement).

Conjecture 1.1 (Transport Conjecture, V2). *Let the universal constants $\sqrt{2\pi e}$, $\sqrt{2/3}$ and the Dijkgraaf–Witten factor $F(N) = (9/10)(N^2 + 1)/N^2$ be as in [R26c]. Let*

$$\mathcal{C}(N) := \sqrt{2\pi e} \cdot \sqrt{\frac{2}{3}} \cdot F(N).$$

For the Bianchi orbifold Y_K with $K = K_{\text{ASP}}(N)$ per Table 1, there exists a transport map $\Phi : \mathcal{O}^{(0^{++})}|_{\mathbb{T}^4} \mapsto m_{\text{spectral}}^{(\chi_)}(K_{\text{ASP}}(N), N)$ such that, in the lattice continuum limit,*

$$\frac{m_{0^{++}}^{\text{lat}}(\text{SU}(N))}{\sqrt{\sigma_N}} = \mathcal{C}(N), \quad (3)$$

where $\sqrt{\sigma_N}$ is the $\text{SU}(N)$ string tension (extracted from the lattice via the standard Wilson loop area-law). Equivalently, the right-hand side of (3) factorises through the spectral observable on Y_K .

1.4. Boolean reformulation (V2 primary falsifier). In the normalisation of [EGM98, §§6.1, 7.4], the continuous spectrum of Δ_{χ} on Y_K starts at $\lambda = 1$. The Selberg-type bound for arithmetic Bianchi 3-manifolds [Sar83] ensures $\lambda_1 \geq 21/25 = 0.84$; an *exceptional* discrete eigenvalue is one in the interval $(21/25, 1)$. Conjecture 1.1 admits the following Boolean reformulation, equivalent to (3) under the universal-factor framing of [R26c]:

Conjecture 1.2 (Boolean Transport, V2 primary falsifier). *For every $N \geq 2$ and the canonical anchor $K = K_{\text{ASP}}(N)$ per Table 1, the lowest cuspidal eigenvalue of Δ_{χ_*} on Y_K is exactly $\lambda_1 = 1$:*

$$\lambda_1(\Delta_{\chi_*}|_{L_{\text{cusp}}^2(Y_K)}) = 1. \quad (4)$$

Equivalently, no exceptional discrete eigenvalue exists in $(21/25, 1)$ on the privileged χ_ -isotypic component.*

Conjectures 1.1 and 1.2 are equivalent under the ECI v16 universal-factor framing where the dimensionless ratio $m_{0++}/\sqrt{\sigma}$ is exhausted by the product $\sqrt{2\pi e} \cdot \sqrt{2/3} \cdot F(N)$. Conjecture 1.2 is operationally sharper: any PARI/GP Selberg-trace eigenvalue search at sufficient precision either finds an exceptional eigenvalue (falsifying both) or confirms its absence (consistent with both).

1.5. Numerical status (V2 update). The Athenodorou–Teper 2021 lattice continuum data [AT21, Tab. 34] for $SU(N)$, $N \in \{2, 3, 4, 5, 6, \infty\}$, gives an RMS deviation of 0.7% from the conjectural value $\mathcal{C}(N)$; see Table 1 of [R26c]. This is consistent with, but does not prove, the conjecture.

Tier (V2): Conjecture 1.1 is TIER 3 OPEN at uniform N . For $N = 3$ specifically, the Boolean reformulation of §1.4 elevates the falsifier to TIER 2 NUM (concrete, budget $\leq \$50$ KVM, ≤ 3 months), pending the PARI Selberg-trace computation.

2. FOUR CANDIDATE PROOF STRATEGIES (V2 RETAINED)

We retain the V1 list of four candidate strategies for promoting Conjecture 1.1 from TIER 3 to TIER 1. Each is treated as a programmatic outline, not a road-mapped proof.

Strategy 1 (CCHS 4D extension). The Chatterjee–Cao–Hairer–Shen programme ([Cha16] and follow-ups; [CCHS20] for the 2D constructive measure; the 4D extension is, to the author’s knowledge, still in progress as of mid-2026; the recent [CNS25] contribution treats the area law in 4D via random surface techniques) provides rigorous lattice $SU(N)$ Wightman observables in 2D and 3D. A 4D extension, when it exists, would provide a rigorous existence theorem for $m_{0++}^{\text{lat}}(SU(N))$ as a Wightman 2-point function correlation. The proof of Conjecture 1.1 via this route would then require: (a) the 4D CCHS construction; (b) identification of the Wightman 2-point function with the spectral observable on a Bianchi orbifold via a Bianchi-cycle gauge fixing.

Status (2026): CCHS-3D published; CCHS-4D in progress (no precise prediction by the author for completion date). The 2025 work of Cao–Nissim–Sheffield [CNS25] on 4D area laws is a positive signal but still does not give the constructive 4D continuum measure. Bianchi-cycle identification: open. Pessimistic 15-year estimate: 10-20%.

Strategy 2 (Cao–Sheffield random surfaces). A programme by Cao–Sheffield (and collaborators) constructs random surface (Liouville) models that, in a certain limit, factor through gauge-theory observables. [CNS25] treats the 4D area law via random surface techniques in lattice gauge theory. If this construction extends to Bianchi 3-orbifolds (manifolds without flat 4D structure), it could provide a Bianchi-to-lattice transport: the Wilson loops on $\mathbb{T}^4$ would arise as random-surface measure averages, and the Bianchi-orbifold spectral observables would be obtained as a homotopy quotient of the Liouville model.

Status (2026): 2D programme partly published; [CNS25] 4D area law published 2025-09. Extension to non-compact Bianchi 3-orbifolds is open.

Strategy 3 (Wightman positivity (W0)). Direct Wightman positivity [SW64]: prove that the lattice 2-point function in (1) is the spectral measure of an appropriate self-adjoint operator on a Hilbert space, and that this operator coincides with Δ_{χ^*} on $L^2(Y_K)_{\chi^*}$ after identification of the gauge-invariant Hilbert space with the Bianchi spectral Hilbert space. This is, in essence, the constructive QFT roadmap for Yang–Mills mass gap, restricted to the surrogate observable.

Status (2026): The full Wightman programme for 4D Yang–Mills is an open Clay Millennium problem [JW00]. A restricted version — proving Wightman positivity *only* for the surrogate observable, not for the full gauge-invariant ensemble — might be more tractable; this is the author’s own preferred long-term route.

Strategy 4 (Cluster decomposition (W5)). Cluster decomposition in the spirit of [Wig56], [HK64]: prove that the lattice 2-point function clusters at large separations, hence its spectral support has a gap. If the spectral gap of the lattice theory coincides with the genus character spectral gap of Δ_{χ^*} on Y_K , the Transport Conjecture would follow at the level of spectral support (though not yet at the level of specific eigenvalue).

Status (2026): Cluster decomposition for lattice $SU(N)$ is folklore-rigorous; the identification with Bianchi spectral support is open.

3. THE TWO-MILLENNIUM-STACK SCOPING (V2 NEW)

The Transport Conjecture is not a single open gap. It is a *2-Millennium-level stack*: the conjecture asserts the existence of a transport map Φ between (M1) the continuum limit of a lattice $SU(N)$ Wightman observable on $\mathbb{T}^4$, and (M2) a spectral observable on Y_K . Each component is independently a Clay-level open problem.

3.1. (M1) Constructive 4D Yang–Mills. The existence of a Wightman quantum field theory for 4D $SU(N)$ Yang–Mills with mass gap is an open Clay Millennium problem [JW00]. No proof of constructive existence is known; the lattice 2-point function (1) is well-defined for any finite lattice spacing a , but the existence of the $a \rightarrow 0$ continuum limit as a Wightman QFT requires:

- proof of reflection positivity for the continuum limit, hence Osterwalder–Schrader reconstruction [SW64];
- proof of the cluster decomposition property [Wig56], [HK64];
- proof of a positive mass gap $m_{0++}^{\text{lat}}(SU(N)) > 0$.

The CCHS programme (Strategy 1) and the Cao–Sheffield programme (Strategy 2) make progress in 2D and 3D; 4D is open.

3.2. (M2) Spectral identification Φ . Assume (M1) is proved. Then *still* the transport map Φ that takes a Wightman observable on $\mathbb{T}^4$ to a spectral observable on Y_K is not known. There is currently no mathematical mechanism that connects a 4-dimensional gauge-theory ensemble to a 3-dimensional arithmetic orbifold spectrum. The numerical 0.7% RMS match (Conjecture 1.1, [R26c]) is suggestive but does not identify the mechanism.

3.3. Joint resolution requires either (a) or (b). A proof of the Transport Conjecture would require one of:

- (a) proof of (M1) via CCHS-4D / Cao–Sheffield / equivalent, plus proof of (M2) as a separate identification step;
- (b) a *joint proof* that bypasses (M1) by identifying the lattice 2-point function with the Bianchi spectral measure directly, without constructing the continuum gauge ensemble first. This would be a Wightman positivity restriction to the surrogate observable (Strategy 3), or a homotopy-categorical bridge (cf. the discussion of [Rw26, Worker 11]).

Both (a) and (b) are wide open. The author’s preferred long-term route is (b), but no candidate path has reached TIER 2 STR maturity.

4. FIVE FALSIFIABLE PREDICTIONS (V2 SHARPENED)

The conjecture yields five testable predictions covering $N \in \{2, 3, 4, 5, 6\}$. Let $\mathcal{C}(N) := \sqrt{2\pi e} \cdot \sqrt{2/3} \cdot F(N)$, and let $K = K_{\text{ASP}}(N)$ per Table 1.

F-T-1 ($N = 2, K = \mathbb{Q}(\sqrt{-15})$). $m_{0++}^{\text{lat}}(SU(2))/\sqrt{\sigma_2} = \mathcal{C}(2) = 3.7962$. *Status (2026)*: AT2021 reports 3.781 ± 0.023 , $\Delta = -0.40\%$, PASS. *Sharper Boolean test*: compute $\lambda_1(\Delta_{\chi_*}|Y_{\mathbb{Q}(\sqrt{-15})})$; predict $\lambda_1 = 1$ (no exceptional eigenvalue) under Conjecture 1.2.

F-T-2 ($N = 3, K = \mathbb{Q}(\sqrt{-67})$). $m_{0++}^{\text{lat}}(SU(3))/\sqrt{\sigma_3} = \mathcal{C}(3) = 3.3744$. *Status (2026)*: AT2021 reports 3.405 ± 0.021 , $\Delta = +0.91\%$, PASS. *Sharper Boolean test*: compute $\lambda_1(\Delta_{\chi_*}|Y_{\mathbb{Q}(\sqrt{-67})})$; predict $\lambda_1 = 1$ (no exceptional eigenvalue) under Conjecture 1.2. **This is the highest-ROI single experiment** for the next phase: PARI/GP Selberg trace formula, $\leq \$50$ KVM, ≤ 3 months.

F-T-3 ($N = 4, K = \mathbb{Q}(\sqrt{-84})$). $m_{0++}^{\text{lat}}(SU(4))/\sqrt{\sigma_4} = \mathcal{C}(4) = 3.2273$. *Status (2026)*: AT2021 reports 3.271 ± 0.027 , $\Delta = +1.34\%$, PASS. *Sharper Boolean test*: compute $\lambda_1(\Delta_{\chi_*}|Y_{\mathbb{Q}(\sqrt{-84})})$ where χ_* is the privileged Klein– V_4 genus character per CR; predict $\lambda_1 = 1$ for that χ_* component.

F-T-4 ($N = 5, K = \mathbb{Q}(\sqrt{-19})$). $m_{0++}^{\text{lat}}(SU(5))/\sqrt{\sigma_5} = \mathcal{C}(5) = 3.1584$. *Status (2026)*: AT2021 reports 3.156 ± 0.031 , $\Delta = -0.06\%$, PASS. *Sharper Boolean test*: compute $\lambda_1(\Delta_{\chi_*}|Y_{\mathbb{Q}(\sqrt{-19})})$ for the trivial character $\chi_* = \chi_0$ (Heegner $h_K = 1$); predict $\lambda_1 = 1$.

F-T-5 ($N = 6$, $K = \mathbb{Q}(\sqrt{-87})$). $m_{0++}^{\text{lat}}(\text{SU}(6))/\sqrt{\sigma_6} = \mathcal{C}(6) = 3.1213$. *Status (2026)*: AT2021 reports 3.102 ± 0.032 , $\Delta = -0.61\%$, PASS.

F-T-6 (universal Boolean, $N \geq 2$). For every $N \geq 2$ and the canonical anchor $K = K_{\text{ASP}}(N)$ per Table 1, no exceptional discrete eigenvalue exists in $(21/25, 1)$ on the privileged χ_* -isotypic component of $L_{\text{cusp}}^2(Y_K)_{\chi_*}$. A single counterexample (an exceptional eigenvalue found at any anchor) falsifies the Transport Conjecture.

The RMS deviation across F-T-1 to F-T-5 is 0.7%, comfortably inside the AT2021 continuum error bars $\sim 3\%$. No datapoint deviates by more than the 1σ AT2021 error bar from $\mathcal{C}(N)$. *The universal Boolean F-T-6 is the cleanest single falsifier* and is the recommended primary test.

5. HONEST OPEN-PROBLEM STATUS (V2 DASHBOARD)

The Transport Conjecture (V2) is an *open problem* with the following tier structure:

- **At uniform N :** TIER 3 OPEN. Conjecture 1.1 is open; 4 candidate strategies (§2) all open; the 2-Millennium-stack (M1) + (M2) scoping (§3) clarifies that any single proof strategy faces two Clay-level walls.
- **At $N = 3$ specifically:** TIER 2 NUM operationally. The Boolean falsifier F-T-2 in §4 is computable in PARI/GP at $\leq \$50$ KVM budget, ≤ 3 months elapsed time.
- **Empirically:** consistent with all known lattice data (AT2021, 0.7% RMS over 6 datapoints). No counterexample known.
- **Strategically:** all four candidate proof strategies in §2 are themselves open programmes. The most tractable near-term option is (b) of §3.3: a Wightman positivity restriction to the surrogate observable.

We invite the community to attempt the conjecture via any of these (or new) strategies.

5.1. Probability estimates (Bayesian, honest). Per the cross-paper analysis [R26f], the Bayesian decomposition is:

- $P(\text{Transport TRUE} \mid \text{empirical RMS } 0.7\%) \approx 65\text{--}80\%$ (suggestive but not conclusive);
- $P(\text{PROVED within 5y} \mid \text{TRUE}) \approx 10\text{--}15\%$ (one candidate path: PARI numerical NUM-tier verification for $\text{SU}(3)$);
- $P(\text{PROVED within 15y} \mid \text{TRUE}) \approx 25\text{--}40\%$;
- $P(\text{PROVED within 30y} \mid \text{TRUE}) \approx 50\text{--}70\%$.

Joint: $P(\text{Clay YM via Transport, 15y}) \approx 0.65 \times 0.30 + 0.80 \times 0.40 = 19.5\text{--}32\%$, median $\approx 26\%$.

5.2. What this paper does *not* claim.

- No proof. The conjecture is stated as an open problem.
- No claim of solution to the Yang–Mills Clay Millennium problem. The conjecture is concerned with a specific surrogate observable, and even at TIER 2 NUM for $N = 3$ it does not establish constructive existence of 4D Yang–Mills.
- No claim that the RMS 0.7% empirical match is a proof. Numerical agreement, however suggestive, is not equivalent to a structural argument.

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